Business Question and Visualization Report

Date	21 October 2026
Team ID	EN25CS3010849
Project Name	Global Olympic Sport Performance Analysis
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations using data stored in Snowflake. Well-framed business questions help identify the required metrics, determine the appropriate Snowflake tables or analytical views, select relevant data, and create visualizations that provide useful business insights.

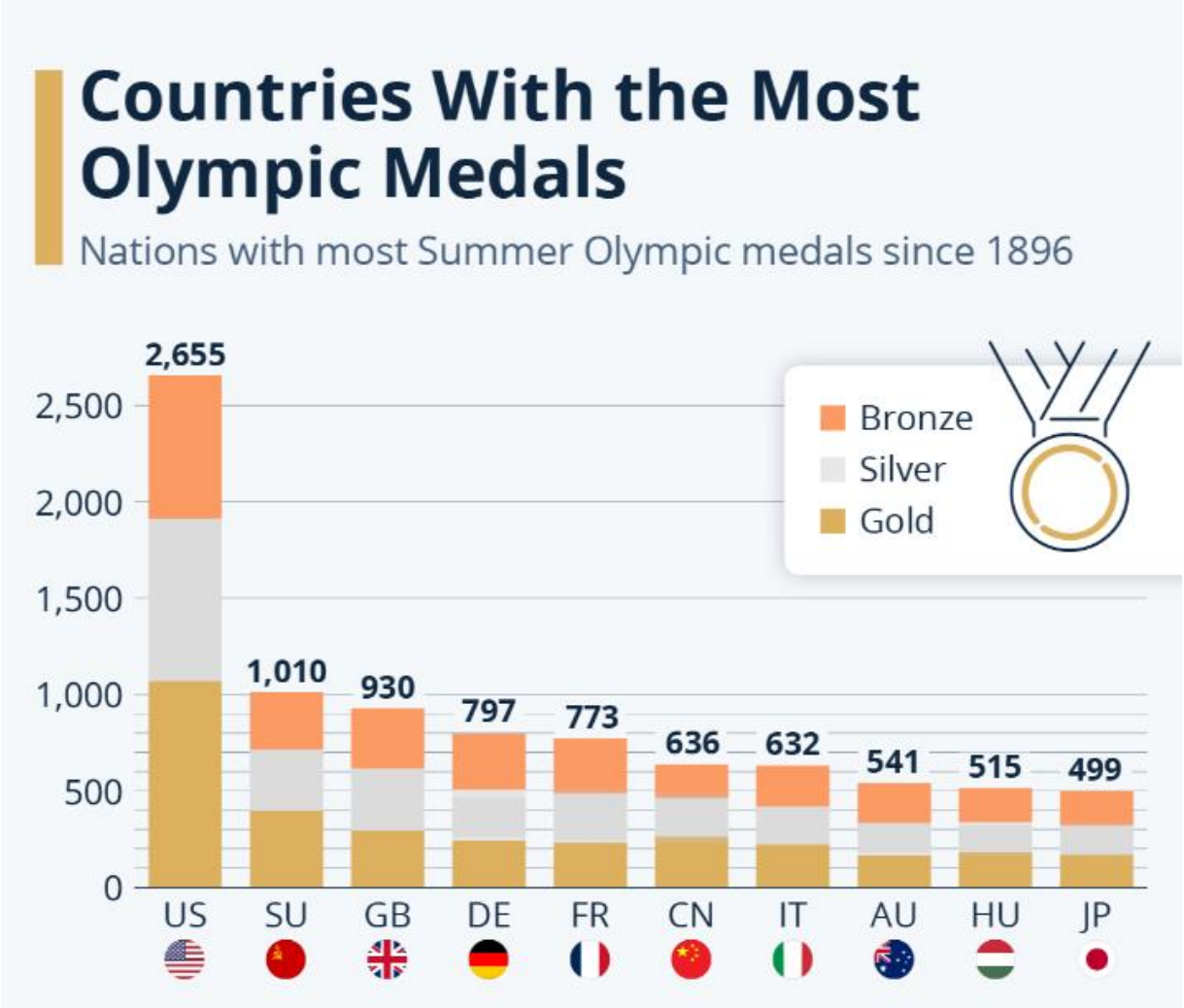
1. What are the top 10 historical gold medal-winning nations in Olympic history?

Code:

```
import pandas as pd

medals_df = pd.read_csv('medals.csv')

gold_lead = (medals_df[medals_df['Medal'] == 'Gold'] .groupby('NOC')['Medal'].count()
.reset_index(name='Gold_Count') .sort_values(by='Gold_Count', ascending=False)
.head(10)) print(gold_lead)
```



2: How has overall athlete participation evolved across summer and winter Olympic years?

- **Description:** Tracks total athlete turnout trends over time across different decades to highlight growth and participation spikes.

Data Extraction Code (Python / SQL):

Python

```
# Python Trend Query for Participation over Years

import pandas as pd

athletes_df = pd.read_csv('athlete_summary.csv')

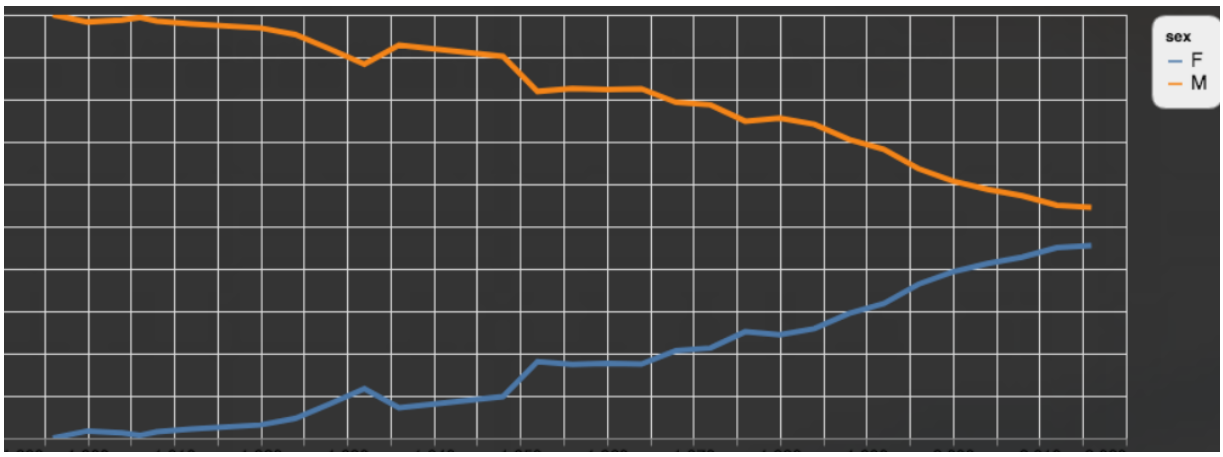
trend_df = (athletes_df.groupby(['Year', 'Season'])['ID']

        .nunique()

        .reset_index(name='Athlete_Count')
```

.sort_values('Year'))

print(trend_df.head(15))



3: How has the ratio of female vs. male Olympic athletes evolved historically?

- **Description:** Evaluates gender inclusivity and historical growth of female participation across Olympic history.

Data Extraction Code (Python / SQL):

Python

Gender Breakdown Query

import pandas as pd

athlete_events = pd.read_csv('athlete.csv')

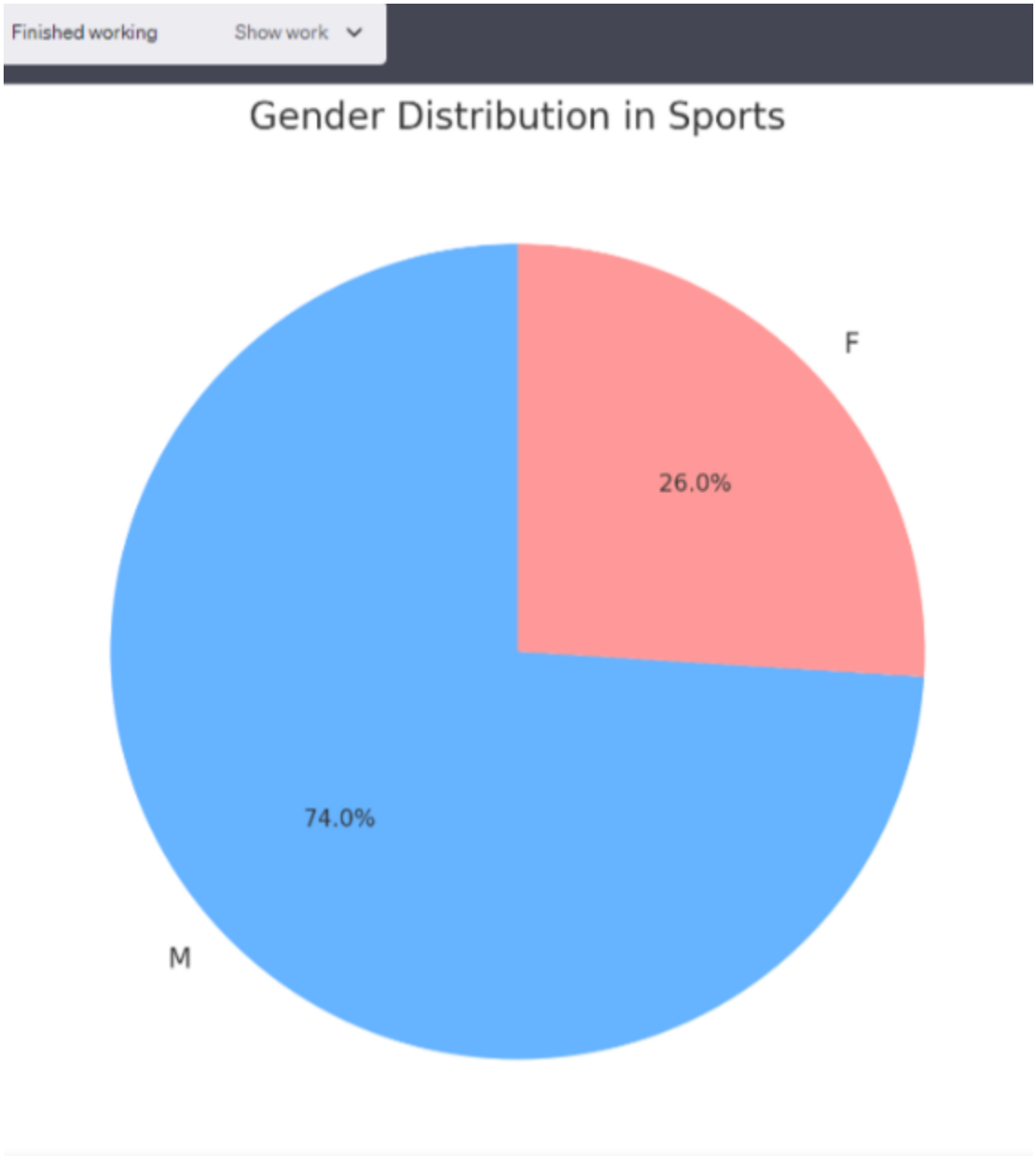
gender_df = (athlete_events.groupby(['Year', 'Sex'])['ID']

.nunique()

.unstack(fill_value=0))

gender_df['Female_Ratio_%'] = (gender_df['F'] / (gender_df['F'] + gender_df['M'])) * 100

print(gender_df.tail(10))



Is there a strong correlation between athlete Height & Weight and medal success in specific sports?

- **Description:** Investigates the physical attribute profiles (Height vs. Weight) of medal-winning athletes across sports like Basketball, Gymnastics, and Athletics.

Data Extraction Code (Python / SQL):

Python

```
# Physical Trait Correlation Code

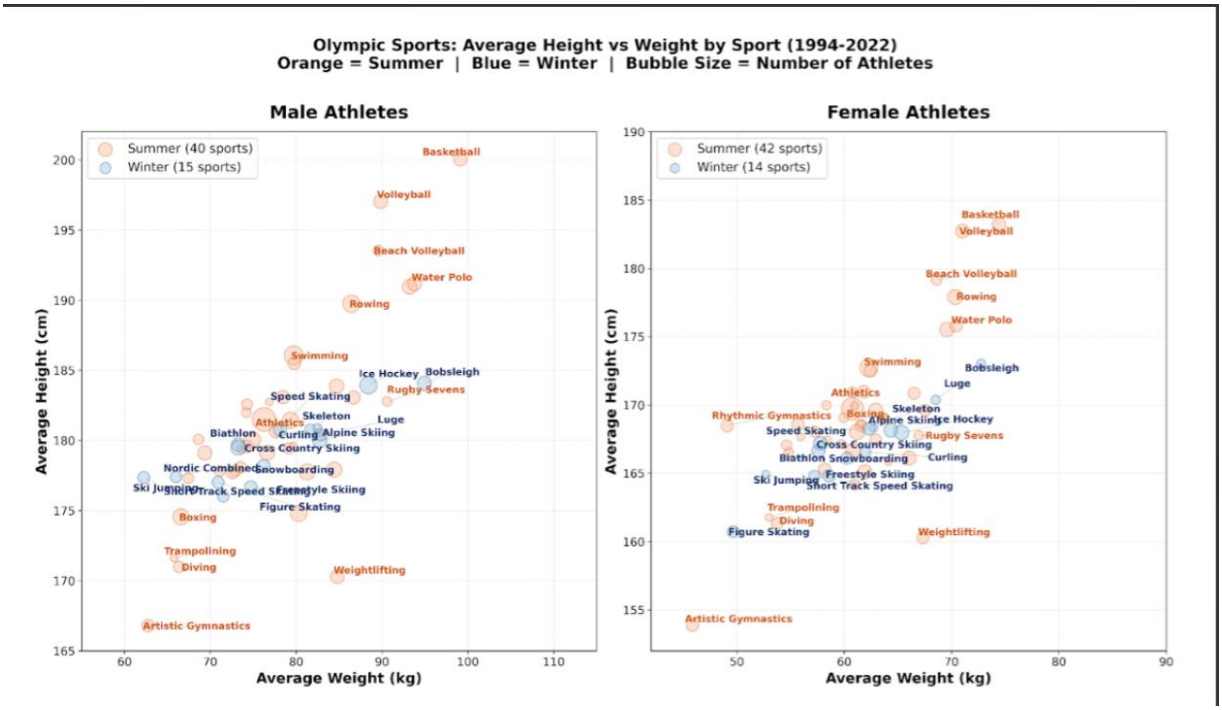
import pandas as pd

athlete_df = pd.read_csv('athlete.csv')

winners = athlete_df[athlete_df['Medal'].notnull()].dropna(subset=['Height', 'Weight'])
```

sports_focus = winners[winners['Sport'].isin(['Basketball', 'Gymnastics', 'Athletics'])]

print(sports_focus[['Name', 'Sport', 'Height', 'Weight', 'Medal']].head(10))



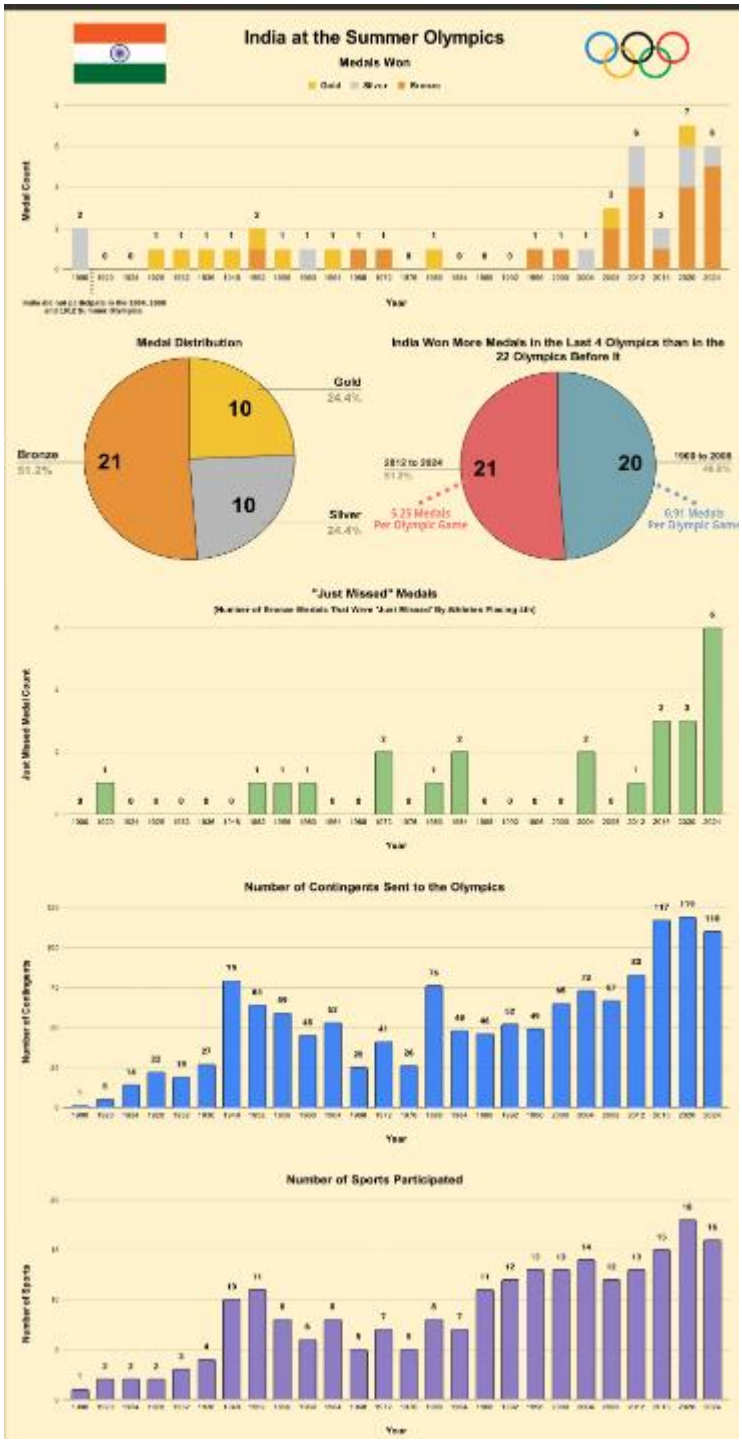
5: Which specific sports disciplines contribute the most medals to India's total tally?

- **Description:** Drills down into India's historical Olympic performance to categorize medal breakdown by sport (e.g., Field Hockey, Shooting, Wrestling).

Data Extraction Code (Python / SQL):

Python

```
# India Medal Distribution Query
import pandas as pd
medals_df = pd.read_csv('medals.csv')
ind_medals = (medals_df[medals_df['NOC'] == 'IND']
        .groupby(['Sport', 'Medal'])['Medal'].count()
        .unstack(fill_value=0))
ind_medals['Total'] = ind_medals.sum(axis=1)
print(ind_medals.sort_values(by='Total', ascending=False))
```



6: What is the age distribution of Olympic medal winners across different sports?

- **Description:** Analyzes the peak competitive age range of medal winners across different event types (e.g., Swimming vs. Equestrian).

Data Extraction Code (Python / SQL):

Python

Age Distribution Query

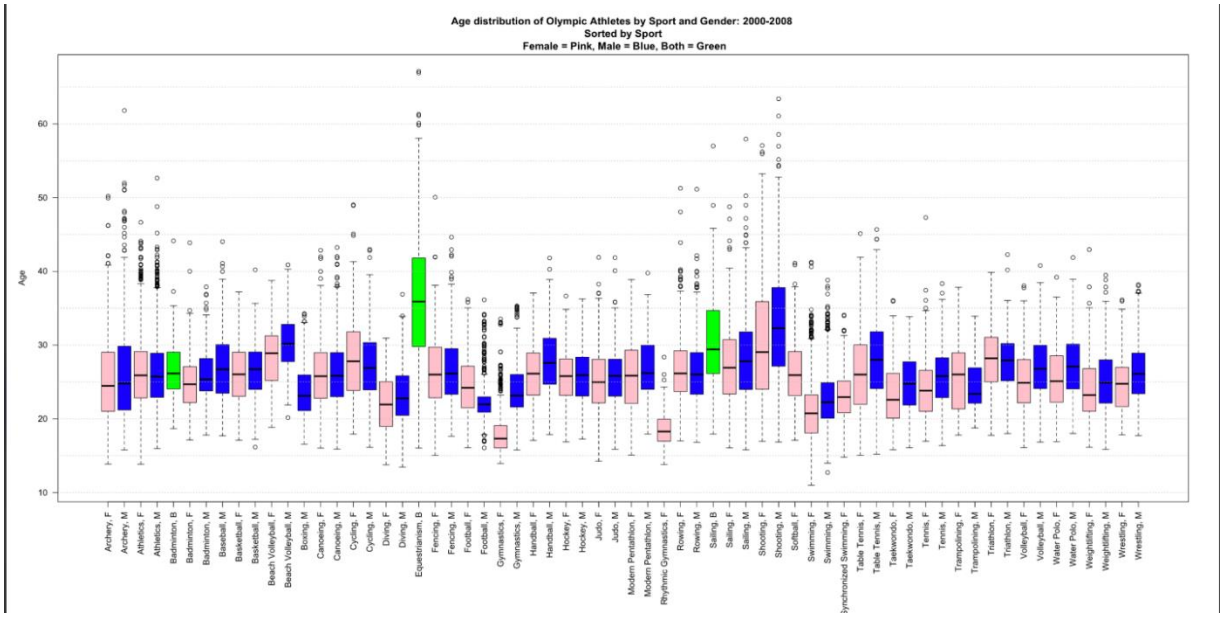
import pandas as pd

athlete_df = pd.read_csv('athlete.csv')

age_df = athlete_df[athlete_df['Medal'].notnull()].dropna(subset=['Age'])

age_summary = age_df.groupby('Sport')['Age'].agg(['min', 'median', 'max']).reset_index()

print(age_summary.head(10))



7: How does overall country performance rank across geographical regions on a global scale?

- **Description:** Provides a global spatial analysis of total Olympic medal concentration across nations.

Data Extraction Code (Python / SQL):

Python

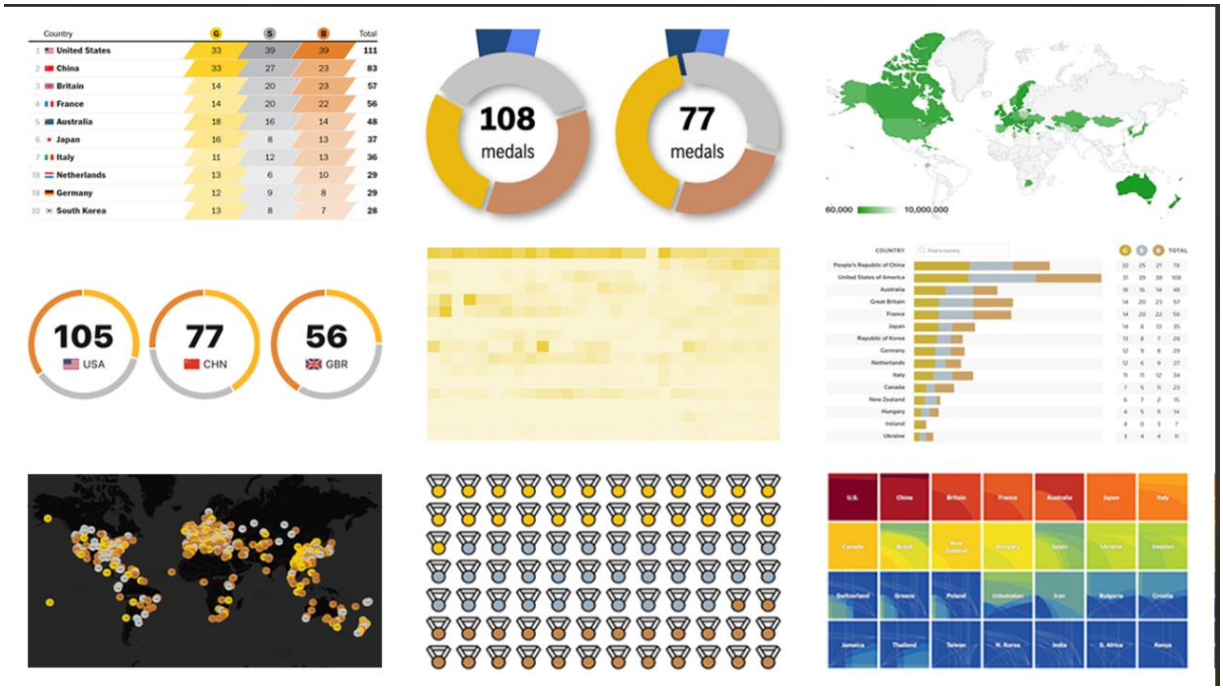
Regional Global Medal Aggregation

import pandas as pd

medals_df = pd.read_csv('medals.csv')

country_tally = medals_df.groupby('NOC')['Medal'].count().reset_index(name='Total_Medals')

print(country_tally.sort_values(by='Total_Medals', ascending=False).head(10))



8: What are the top 10 individual athletes with the highest total medal counts in Olympic history?

- **Description:** Identifies legendary individual Olympic athletes across all eras and ranks them by total career medals.

Data Extraction Code (Python / SQL):

Python

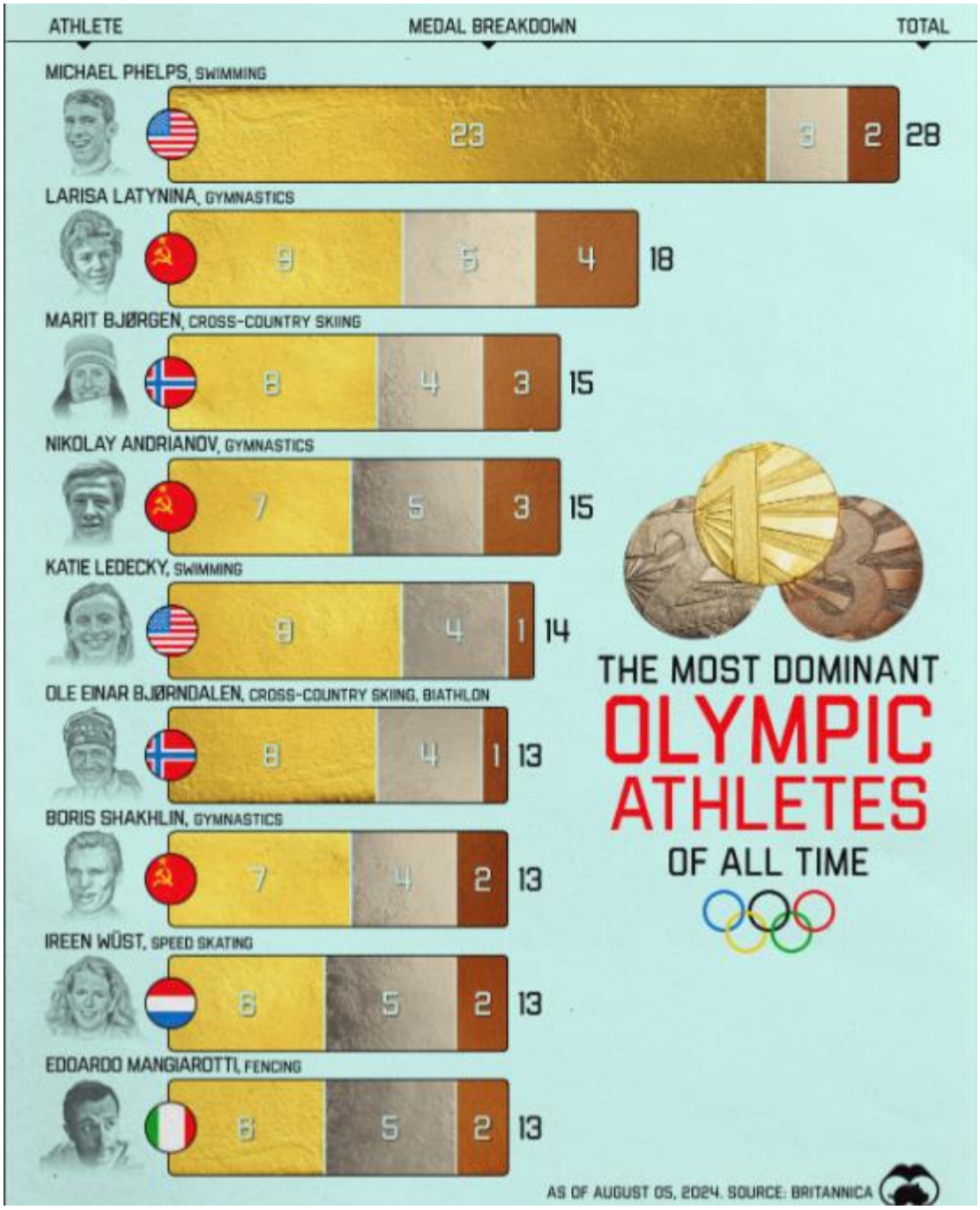
```
# Top Individual Olympians Query

import pandas as pd

athlete_events = pd.read_csv('athlete.csv')

top_athletes = (athlete_events[athlete_events['Medal'].notnull()]
        .groupby(['Name', 'NOC', 'Sport'])['Medal'].count()
        .reset_index(name='Total_Medals')
        .sort_values(by='Total_Medals', ascending=False)
        .head(10))

print(top_athletes)
```


ATHLETE
MEDAL BREAKDOWN
TOTAL
MICHAEL PHELPS, SWIMMING
23
3
2
28
LARISA LATYNINA, GYMNASTICS
9
5
4
18
MARIT BJØRGEN, CROSS-COUNTRY SKIING
8
4
3
15
NIKOLAY ANDRIANOV, GYMNASTICS
7
5
3
15
KATIE LEDECKY, SWIMMING
9
4
1
14
OLE EINAR BJØRNDALEN, CROSS-COUNTRY SKIING, BIATHLON
8
4
1
13
BORIS SHAKHLIN, GYMNASTICS
7
4
2
13
IREEN WÜST, SPEED SKATING
6
5
2
13
EDOARDO MANGIAROTTI, FENCING
6
5
2
13
THE MOST DOMINANT
OLYMPIC
ATHLETES
OF ALL TIME
AS OF AUGUST 05, 2024. SOURCE: BRITANNICA